# Capstone Project

## E-commerce Customer behavior Analysis

From customer behavior and product reviews to analyze the Lifetime Value (LTV) and experience the business insights

**Group 8:**
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**50k**

Customer-level rows in the source CSV

**25**

Original columns across profile, behavior, value, and review text

**17**

Python charts generated from the analysis workflow

# Project Objective

**Business question**

## What drives customer value?

Identify which customer behaviors, engagement signals, and review sentiment patterns are most associated with Lifetime Value.

**Decision support**

## Where should the business act?

Turn EDA and NLP outputs into targeting ideas for retention, conversion, and experience improvement.

# Dataset Source from Kaggle

| Column group | Fields | How it is used |
|---|---|---|
| **Profile** | Age, Gender, Country, City | Segmentation and audience context |
| **Activity** | Membership_Years, Login_Frequency, Session_Duration_Avg, Pages_Per_Session | Engagement intensity and browsing depth |
| **Purchase value** | Total_Purchases, Average_Order_Value, Lifetime_Value | Customer value and monetization |
| **Friction and service** | Cart_Abandonment_Rate, Returns_Rate, Customer_Service_Calls | Experience risk and operational drag |
| **Marketing engagement** | Email_Open_Rate, Discount_Usage_Rate, Wishlist_Items | Campaign response and intent |
| **Text and labels** | Reviews, Churned, Signup_Quarter | NLP input and optional business labels |

Kaggle: https://www.kaggle.com/datasets/dhairyajeetsingh/ecommerce-customer-behavior-dataset/data

# Main Columns And Data Categories

These columns carry the main analytical meaning in the deck. Profile fields describe who the customer is, behavior fields explain how they shop, value fields define business outcomes, and review text supports sentiment analysis.

| Data category | Main columns | Data type | Business use |
|---|---|---|---|
| **Customer profile** | Age, Gender, Country, City | Numeric + categorical | Segment customers and compare audience context |
| **Tenure** | Membership_Years | Numeric | Understand customer maturity and lifecycle stage |
| **Engagement behavior** | Login_Frequency, Session_Duration_Avg, Pages_Per_Session | Numeric | Measure shopping depth and activity intensity |
| **Purchase value** | Total_Purchases, Average_Order_Value, Lifetime_Value | Numeric | Define monetization and the main LTV outcome |
| **Conversion friction** | Cart_Abandonment_Rate, Days_Since_Last_Purchase | Numeric rates + recency | Identify checkout and reactivation risk |
| **Retention risk** | Returns_Rate, Customer_Service_Calls | Numeric | Capture service burden and post-purchase friction |
| **Marketing response** | Email_Open_Rate, Discount_Usage_Rate | Numeric rates | Evaluate campaign response and price sensitivity |
| **Purchase intent** | Wishlist_Items, Payment_Method_Diversity | Numeric | Spot buying intent and payment behavior breadth |
| **Review text** | Reviews | Text | Input for VADER sentiment scoring and issue mining |

# Missingness Is Manageable But Not Randomly Ignorable


## What it shows

- Highest missingness is in social engagement, credit balance, and mobile usage.
- Notebook cleaning removes several high-missing or non-core fields before EDA.
- Remaining numeric missing values are filled with medians, which stabilizes charts but can compress variance.

```
missing_summary = pd.DataFrame({
    'missing_count': df.isnull().sum(),
    'missing_percentage': df.isnull().mean() * 100
})

missing_summary = missing_summary.sort_values(
    by='missing_percentage',
    ascending=False
)

missing_summary['missing_percentage'] = missing_summary['missing_percentage'].round(2).as

missing_summary
```

| ... | # missing_count | missing_percentage |
|---|---|---|
| Social_Media_Engagement_Score | 6000 | 12.0% |
| Credit_Balance | 5500 | 11.0% |
| Mobile_App_Usage | 5000 | 10.0% |
| Returns_Rate | 4491 | 8.98% |
| Wishlist_Items | 4000 | 8.0% |
| Discount_Usage_Rate | 3500 | 7.0% |
| Session_Duration_Avg | 3399 | 6.8% |
| Days_Since_Last_Purchase | 3000 | 6.0% |
| Pages_Per_Session | 3000 | 6.0% |
| Email_Open_Rate | 2528 | 5.06% |
| Payment_Method_Diversity | 2500 | 5.0% |
| Age | 2495 | 4.99% |

# Data Processing Decisions

Column removal

## High-missing and out-of-scope fields were dropped

Social_Media_Engagement_Score, Mobile_App_Usage, Credit_Balance, Churned, and Signup_Quarter were removed for this deck's EDA path.

```
df = df.drop([
    'Social_Media_Engagement_Score',
    'Mobile_App_Usage',
    'Credit_Balance',
    'Mobile_App_Usage',
    'Churned',
    'Signup_Quarter'
    ], axis=1)
df
```

Imputation

## Numeric gaps were median-filled

Median imputation keeps all 50,000 rows available. It is robust to outliers but can weaken true segment differences.

```
df[num_cols] = df[num_cols].fillna(df[num_cols].median())

print(df[num_cols].isna().sum())
```

```
Age                         0
Membership_Years            0
Login_Frequency             0
Session_Duration_Avg        0
Pages_Per_Session           0
Cart_Abandonment_Rate       0
Wishlist_Items              0
Total_Purchases             0
Average_Order_Value         0
Days_Since_Last_Purchase    0
Discount_Usage_Rate         0
Returns_Rate                0
Email_Open_Rate             0
Customer_Service_Calls      0
Payment_Method_Diversity    0
Lifetime_Value              0
dtype: int64
```

Validation

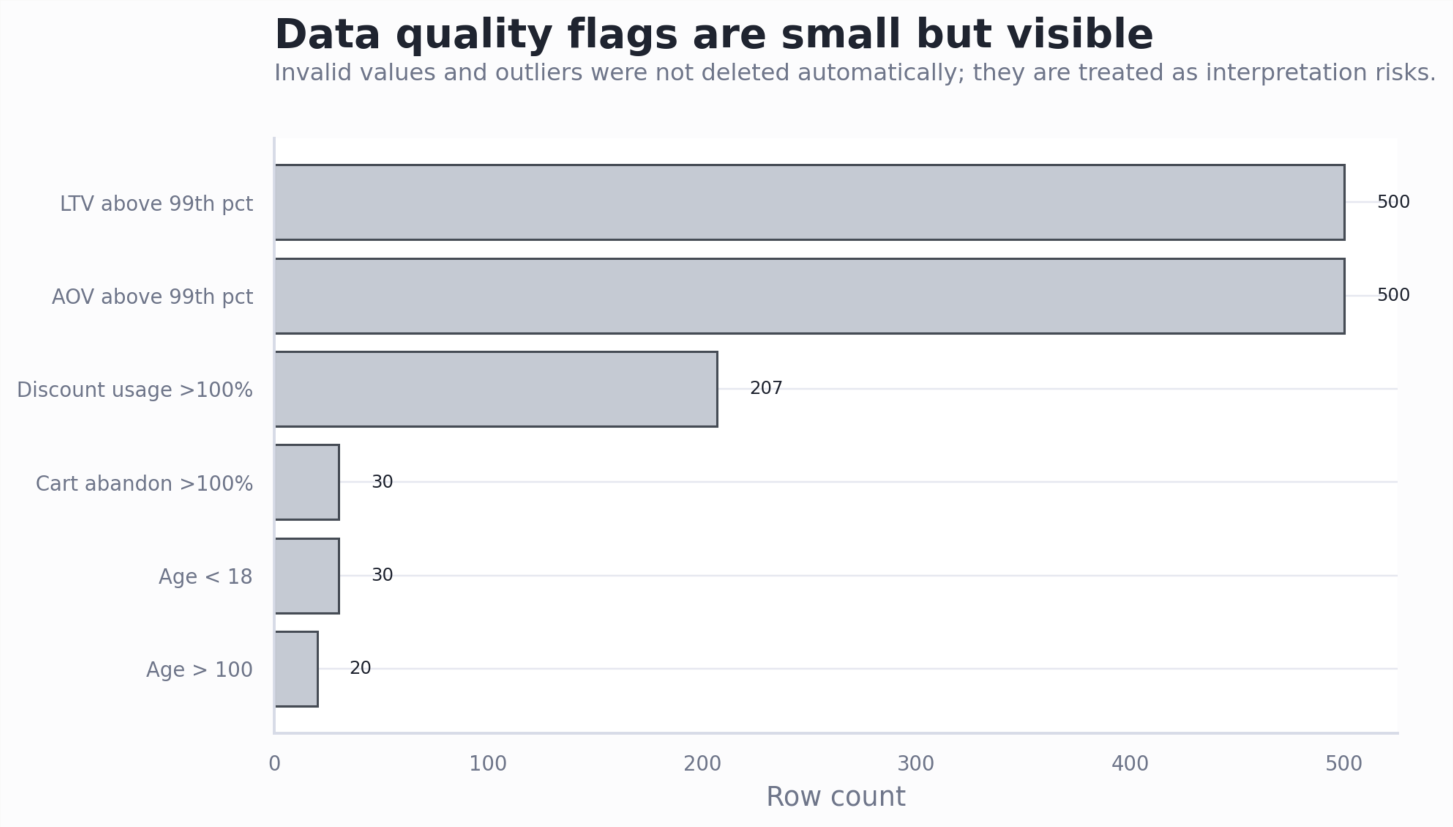
## Duplicates and impossible values were checked

No full duplicate rows were found. A small number of invalid ages and rate values above 100% were flagged as limitations.

```
total_duplicates = df.duplicated().sum()
print(f"Duplicate row: {total_duplicates}")
```

```
Duplicate row: 0
```

# Quality Flags Are Small Enough To Analyze, But They Matter




**What it shows**

- AOV and LTV have long-tail outliers; most charts trim top 1% only for readability.
- Age includes values under 18 and above 100.
- Some rate fields exceed 100%, which should be repaired before production modeling.

**500**

AOV rows above the 99th percentile

**237**

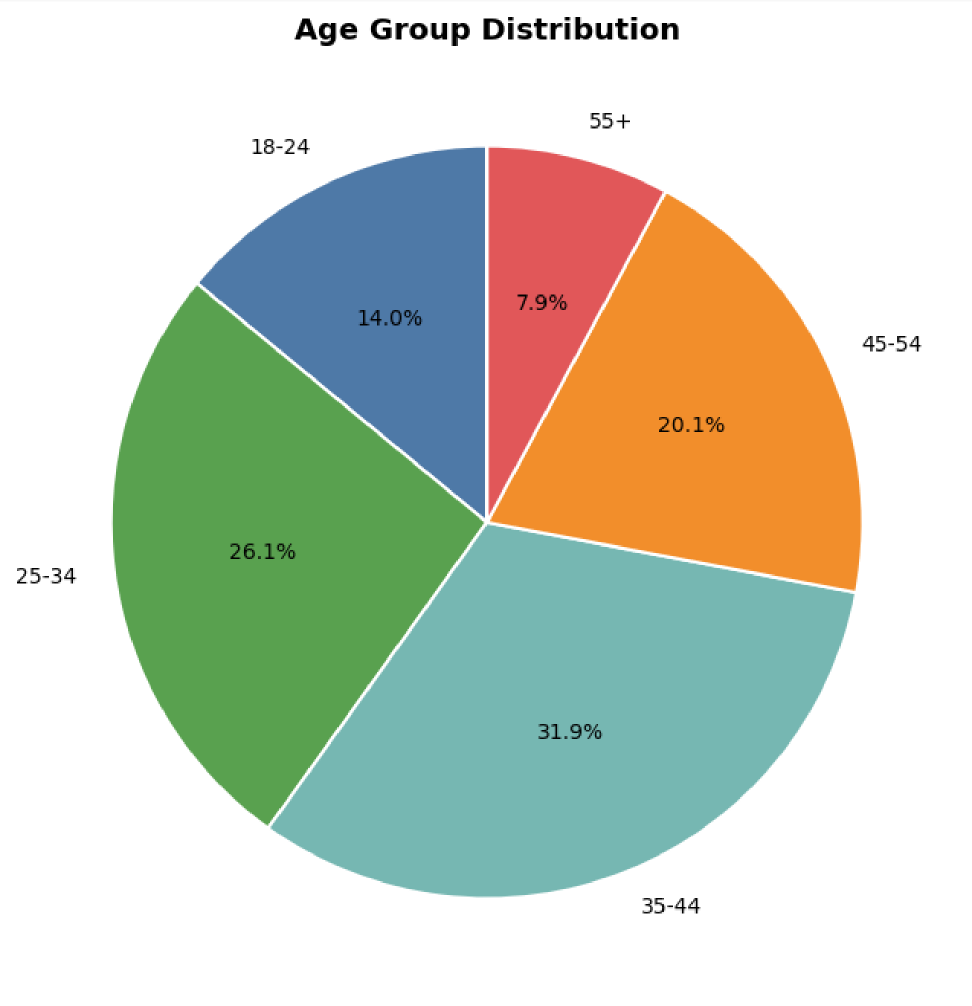
Rows with cart or discount rates above 100%

# EDA Analysis

The EDA moves from distributions to relationships, then to multi-signal customer segments.

- Univariate analysis
- Bivariate analysis
- Multivariate analysis

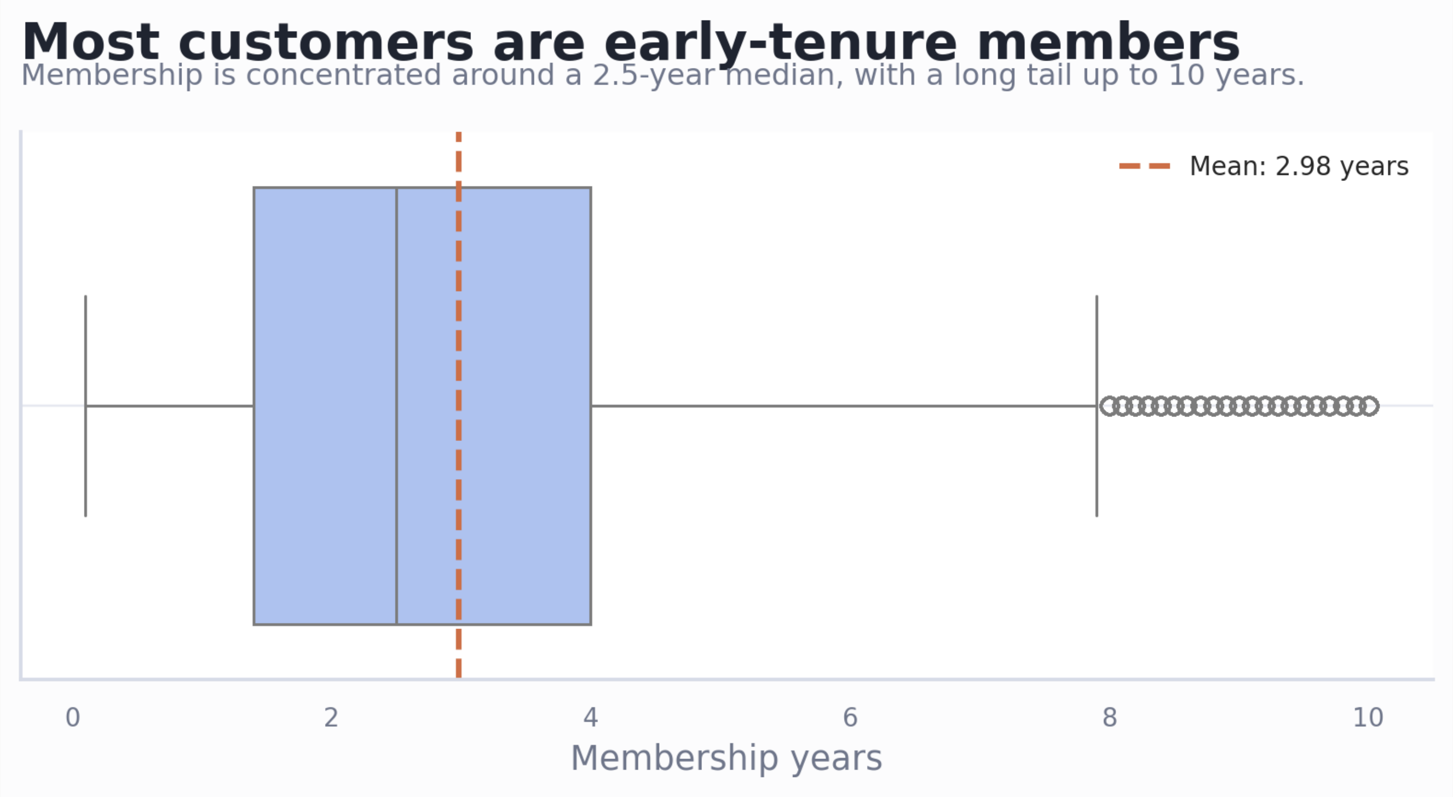
# Univariate: Age Group




**What it shows**

- The 35-44 group with 31.9% is the largest customer portion in this dataset.
- Age alone is useful for audience context, but it does not explain LTV strongly.

# Univariate: Membership Years




**What it shows**

- Median membership tenure is 2.5 years, while the mean is about 3.0 years.
- The tenure distribution suggests many customers are still in an activation and growth window.
- Outlier values appear at the high-tenure edge but are within the 10-year cap.

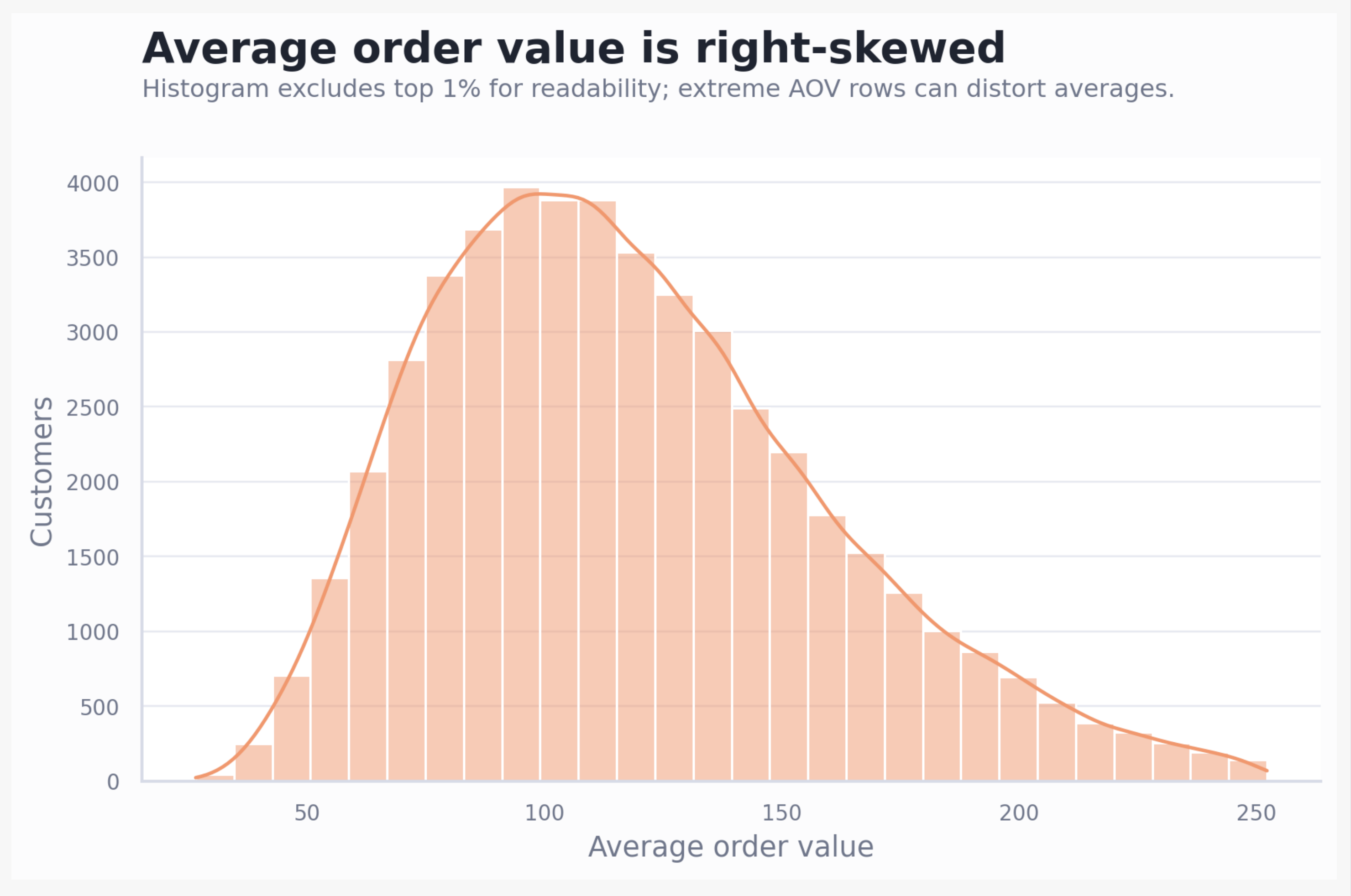
**2.5y**

Median membership years

**3.0y**

Mean membership years

# Univariate: AOV is right-skewed




**What it shows**

- Most average order values cluster around the low-to-mid range.
- The chart trims the top 1% for readability because the raw maximum is extreme.
- Business decisions should compare median and percentile bands, not only average AOV.

**$113**

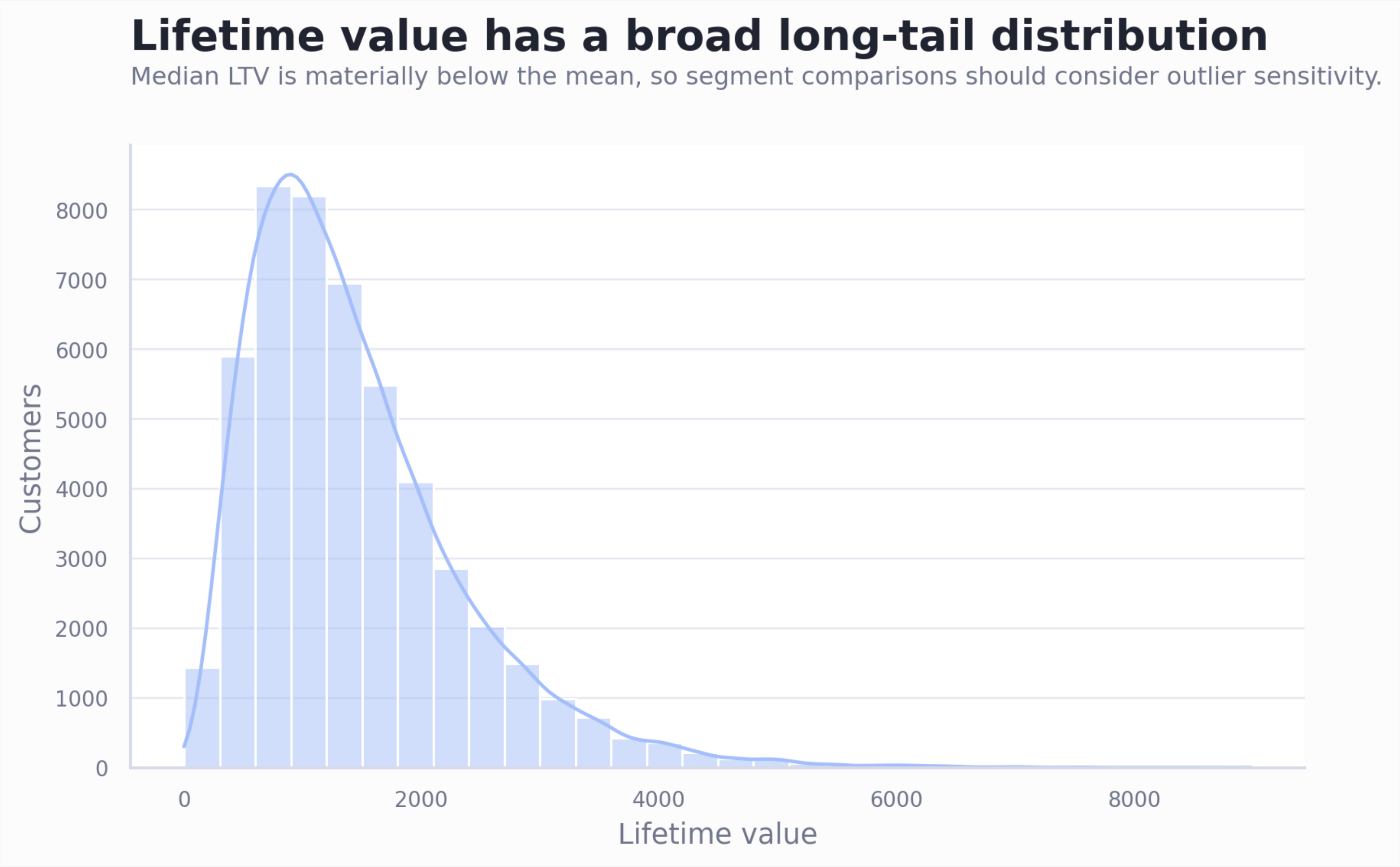
Median AOV after cleaning

**$123**

Mean AOV after cleaning

ANALYSIS EVIDENCE

# Univariate: LTV Has A Long Tail Of High-Value Customers




**What it shows**

- Most customers sit below the mean LTV, while a smaller group contributes very high value.
- This shape supports targeted retention and upsell strategies rather than one-size-fits-all messaging.
- Outliers can distort segment averages, so median and percentile checks should accompany mean comparisons.

**$1.24k**
Median LTV

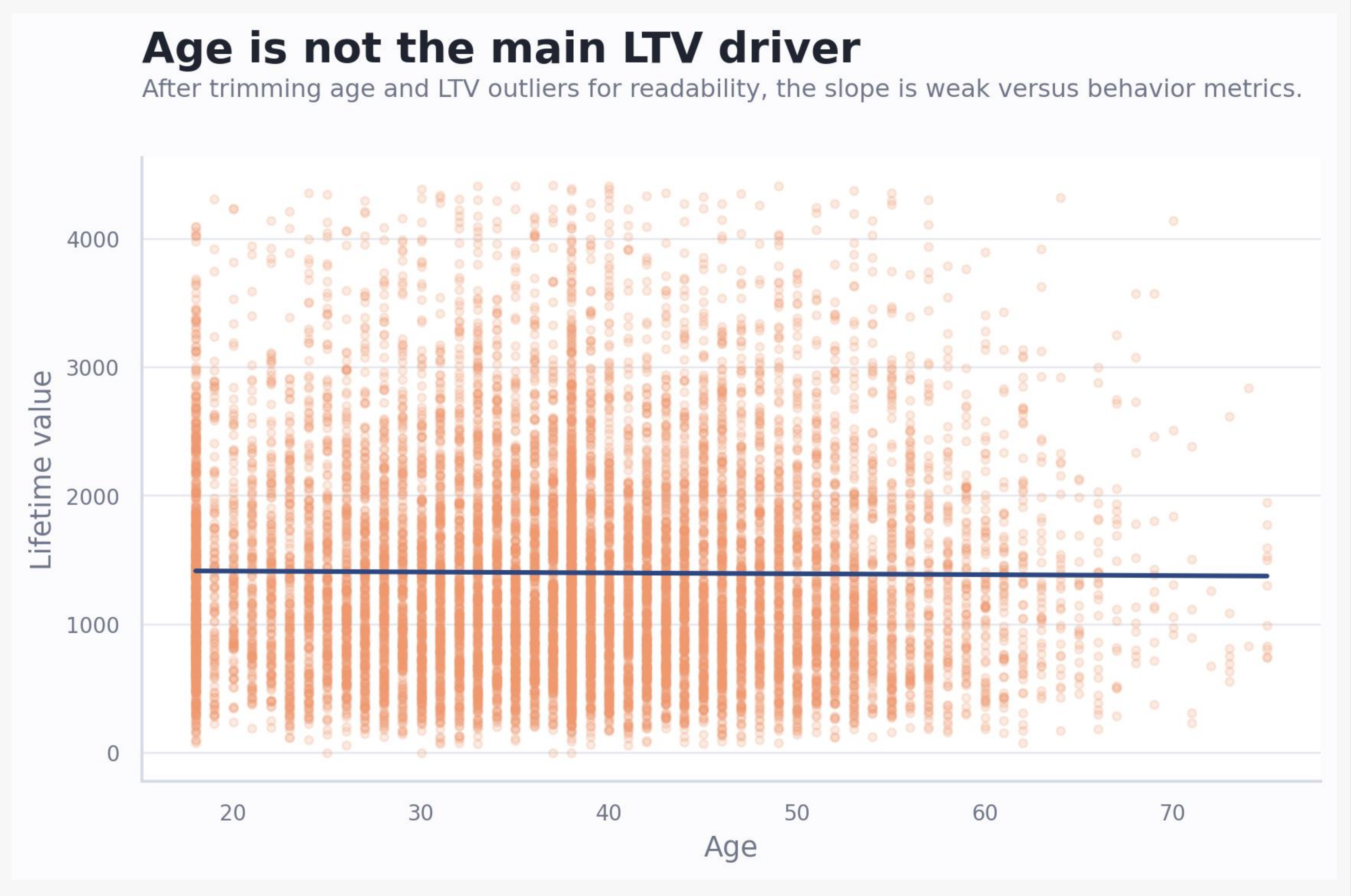
**$1.44k**
Mean LTV

# Bivariate Analysis

- Demographics vs LTV
- Engagement vs LTV
- Friction vs LTV

Pairwise relationships identify which variables deserve business attention and which are mostly context.

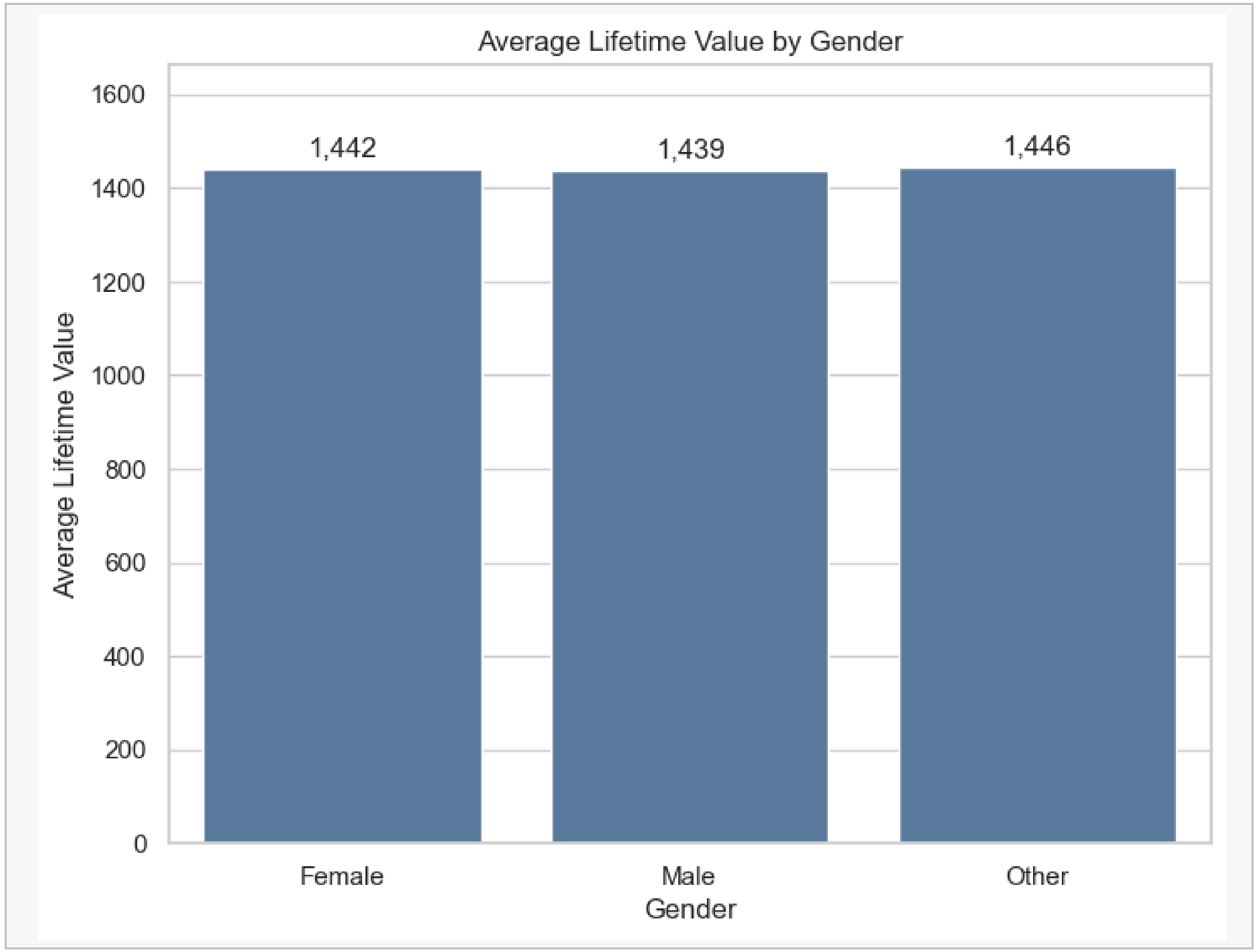
# Bivariate: Age Does Not Explain LTV Strongly




**What it shows**

- The fitted line is nearly flat compared with behavior-based relationships.
- Age should not be the primary basis for value targeting in this dataset.
- Use age as campaign context only after validating age data quality.

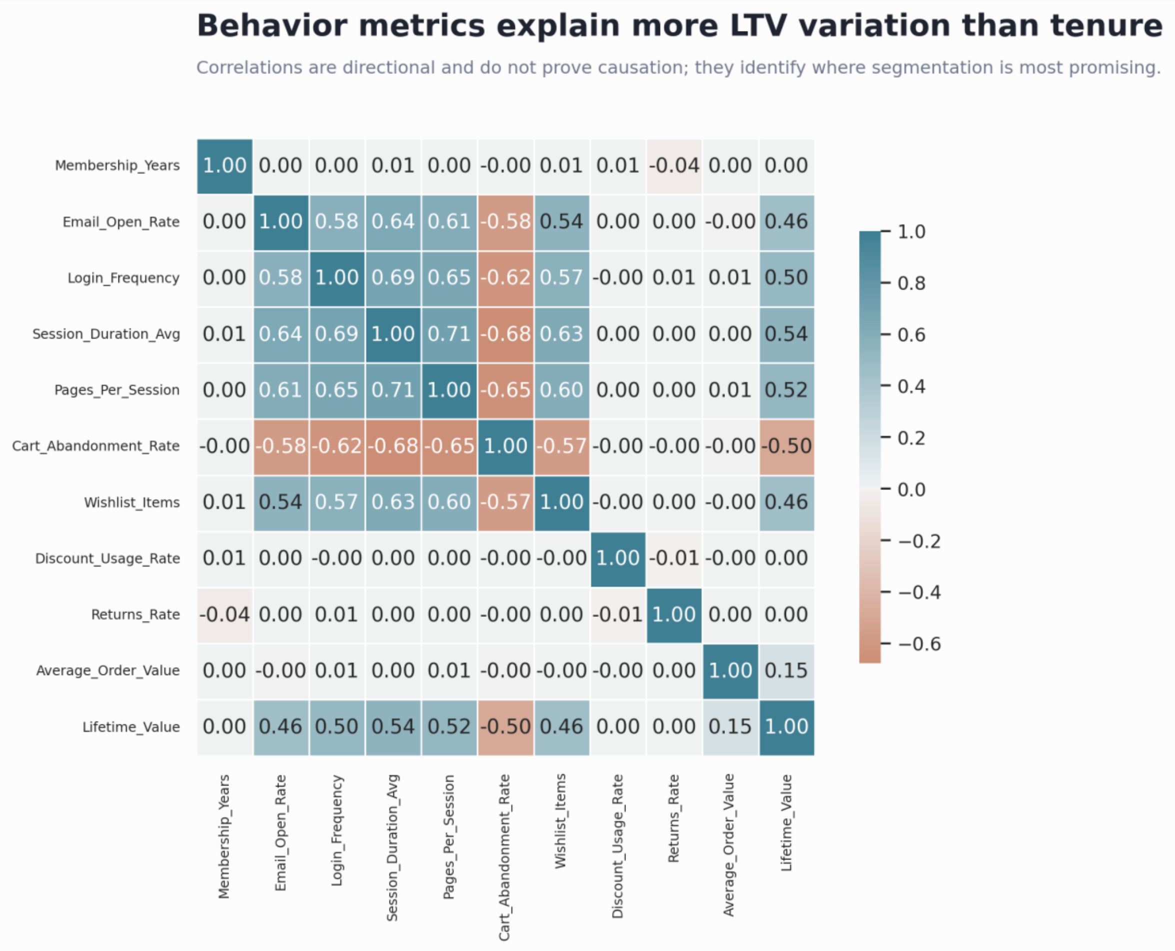
# Bivariate: Gender Does Not Explain LTV Strongly




**What it shows**

- Gender should not be the primary basis for value targeting in this dataset.
- Use Gender as campaign context only after validating age data quality.

# Bivariate: Correlations Point To Behavior, Not Demographics




### What it shows

- Total purchases, session duration, pages per session, login frequency, and email open rate move with LTV.
- Cart abandonment is negatively associated with LTV and likely marks conversion friction.
- Correlation is not causation, but this map guides which segments deserve deeper modeling.

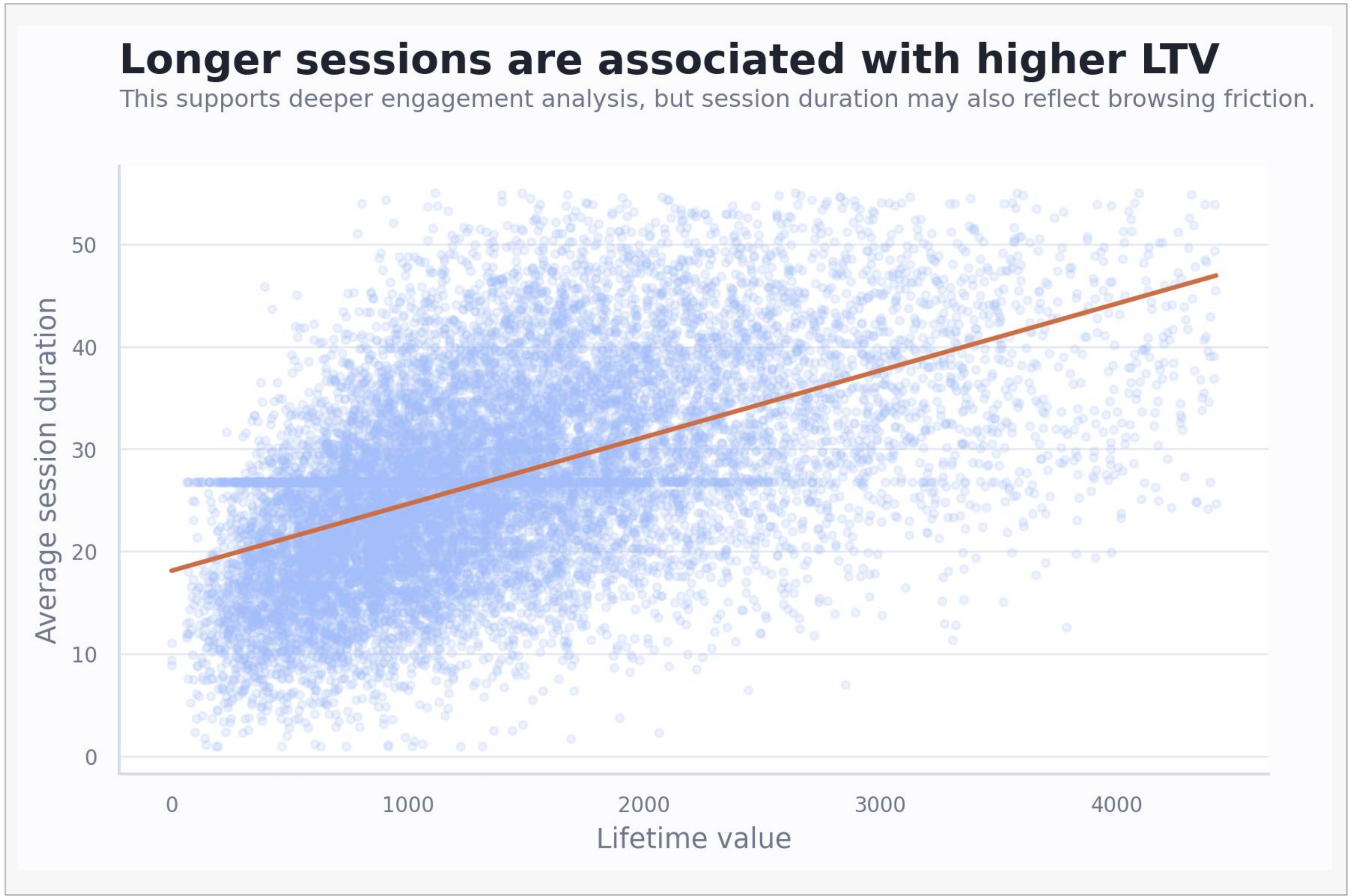
**0.62**

Total Purchases to LTV correlation

**-0.50**

Cart Abandonment to LTV correlation

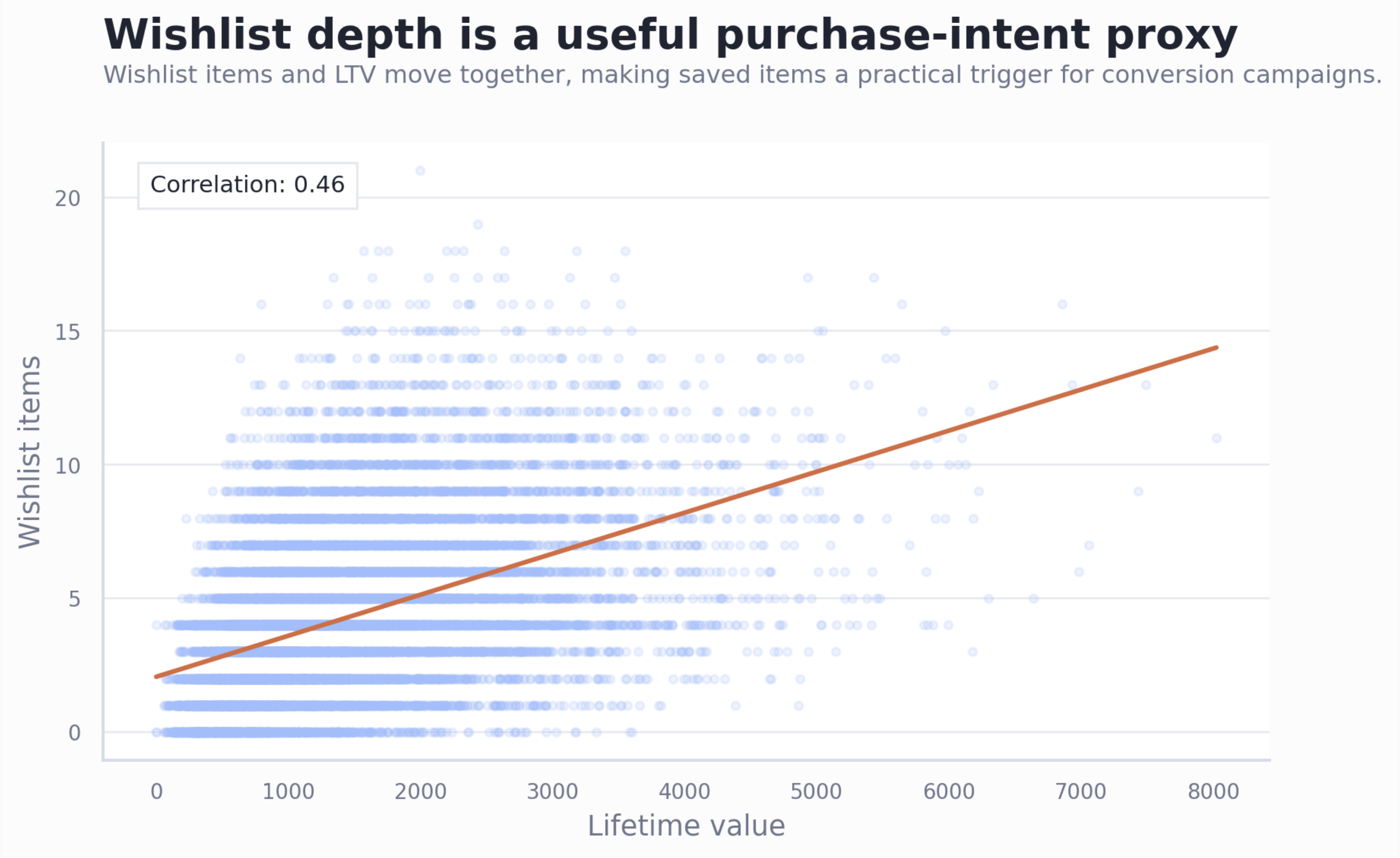
# Bivariate: Session Depth Is Associated With Higher LTV




**What it shows**

- Higher LTV customers tend to have longer average sessions.
- This may reflect stronger engagement, but it could also indicate browsing friction for some users.
- Next analysis should split long sessions by conversion, abandonment, and review sentiment.

# Bivariate: Wishlist Items Are A Practical Intent Signal




**What it shows**

- Wishlist count and LTV have a moderate positive relationship.
- Saved items can support triggered reminders, limited-time offers, and personalized recommendations.
- The signal should be tested against conversion lift before treating it as causal.

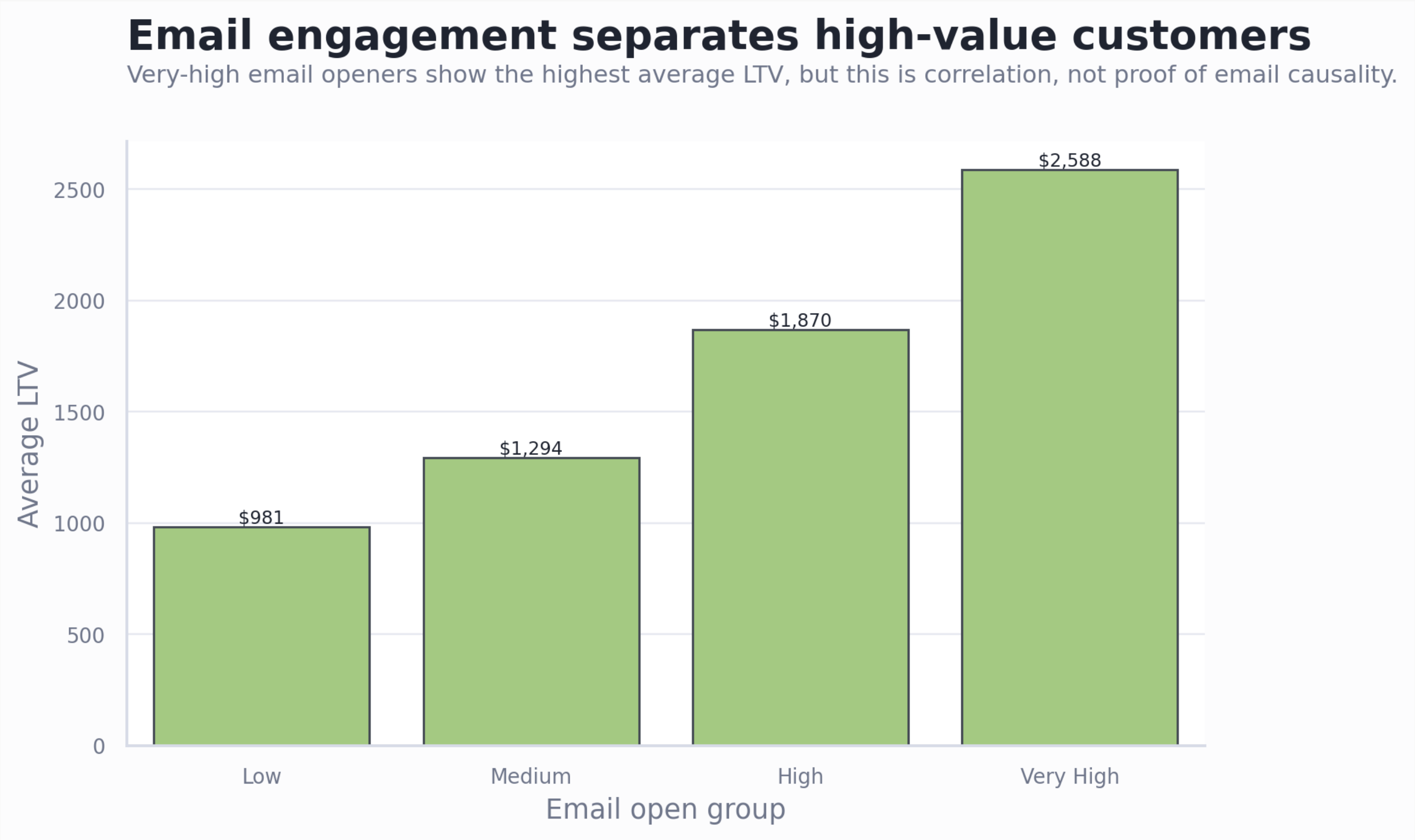
**0.46**

Wishlist to LTV correlation

**Intent**

Useful for targeting and lifecycle nudges

# Bivariate: Email Engagement Separates Higher-Value Customers




**What it shows**

- Average LTV rises across email open groups, peaking in the very-high open segment.
- Email engagement may be a strong prioritization signal for campaigns and retention plays.
- This does not prove email causes LTV; high-value customers may simply open more messages.

# Multivariate Analysis

Combining signals helps separate high-value intent from friction and avoids over-reading one variable at a time.

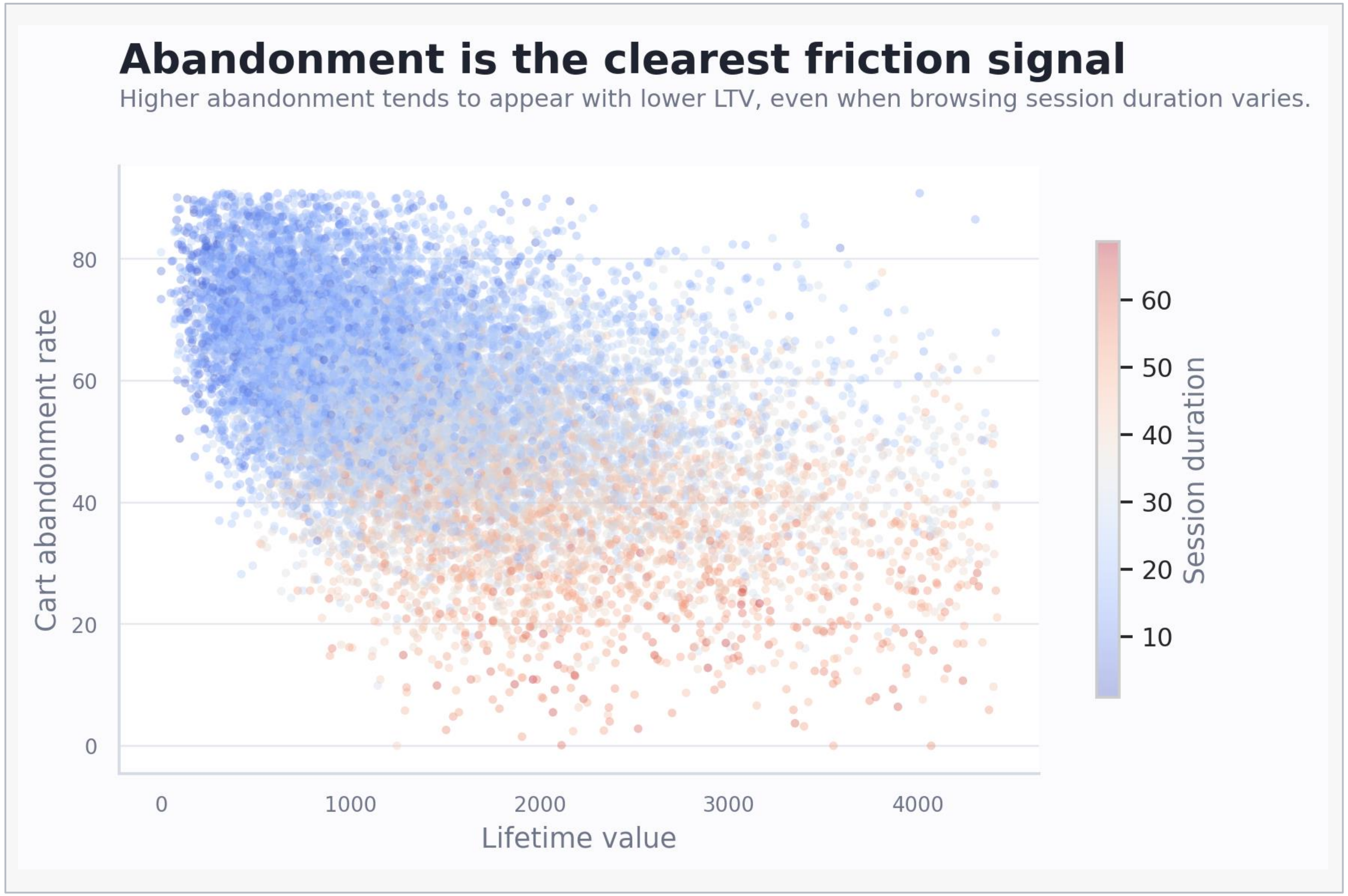
- Engagement plus intent
- Friction plus session depth
- Tenure plus purchase frequency

# Multivariate: High-LTV Customers Combine Engagement And Intent


**What it shows**

- The strongest value region combines higher LTV, longer sessions, and more wishlist items.
- This is a practical segment for personalized recommendations and wishlist conversion journeys.
- The chart is sampled for readability, so final targeting should score the full dataset.

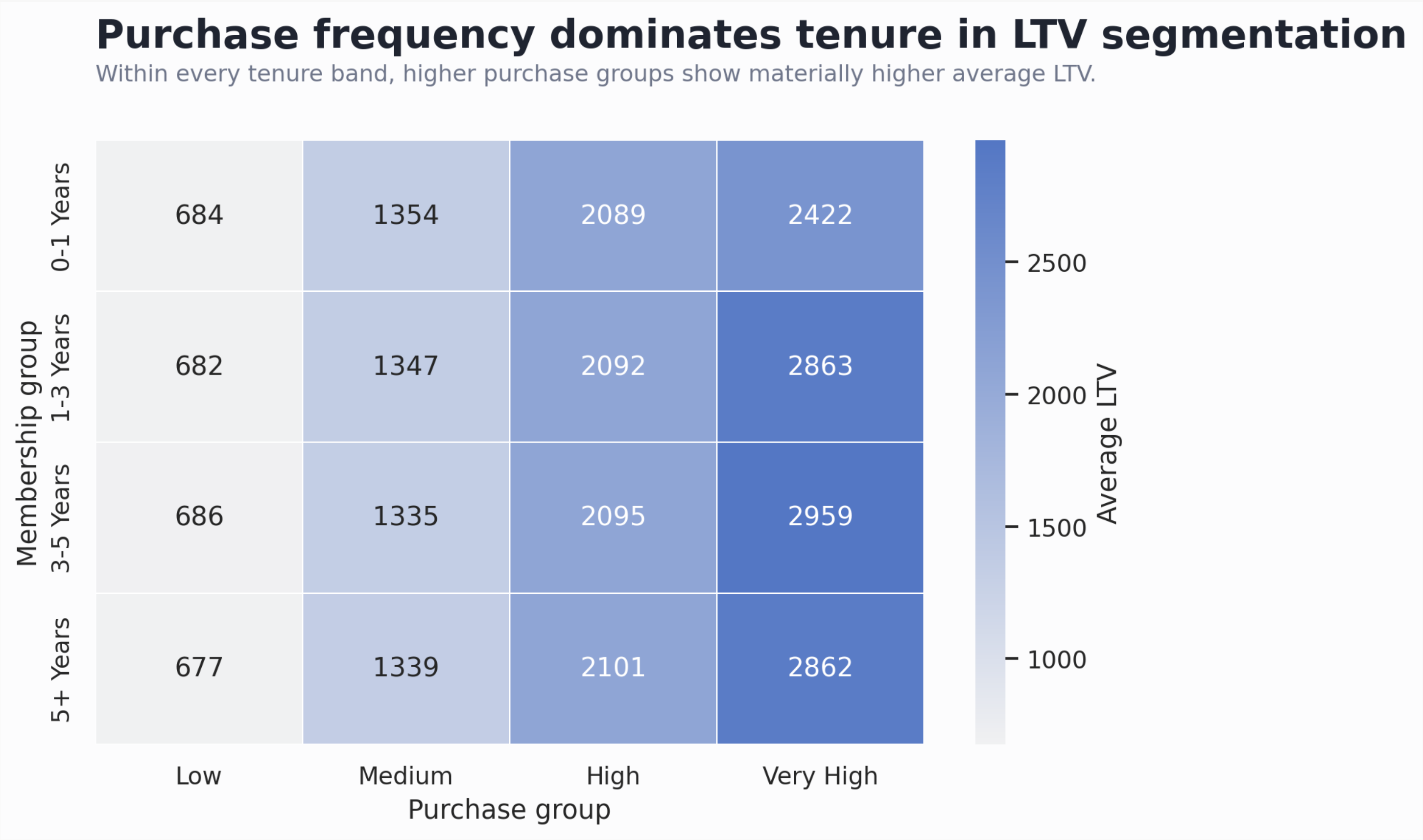
# Multivariate: Abandonment Is The Clearest Friction Signal




**What it shows**

- Higher abandonment rates are more common in lower-LTV regions.
- Long sessions are not automatically good when they come with high cart abandonment.
- Checkout and returns diagnostics should focus on high-intent users who still abandon carts.

# Multivariate: Purchase Frequency Dominates Tenure




**What it shows**

- Within every membership year, higher purchase groups have much higher average LTV.
- Tenure alone is not enough to identify customer value.
- Lifecycle strategy should combine tenure with purchase frequency, not treat older accounts as automatically valuable.

# Machine Learning: Sentiment Analysis

Review text adds a customer-experience layer that numeric behavior alone cannot show.

- VADER compound scoring
- Positive, neutral, negative labels
- Sentiment compared with value

# Sentiment Model Setup

Input

### Customer review text

Each review was scored using VADER sentiment intensity. The compound score summarizes polarity from negative to positive.

Label rule

### Threshold-based classification

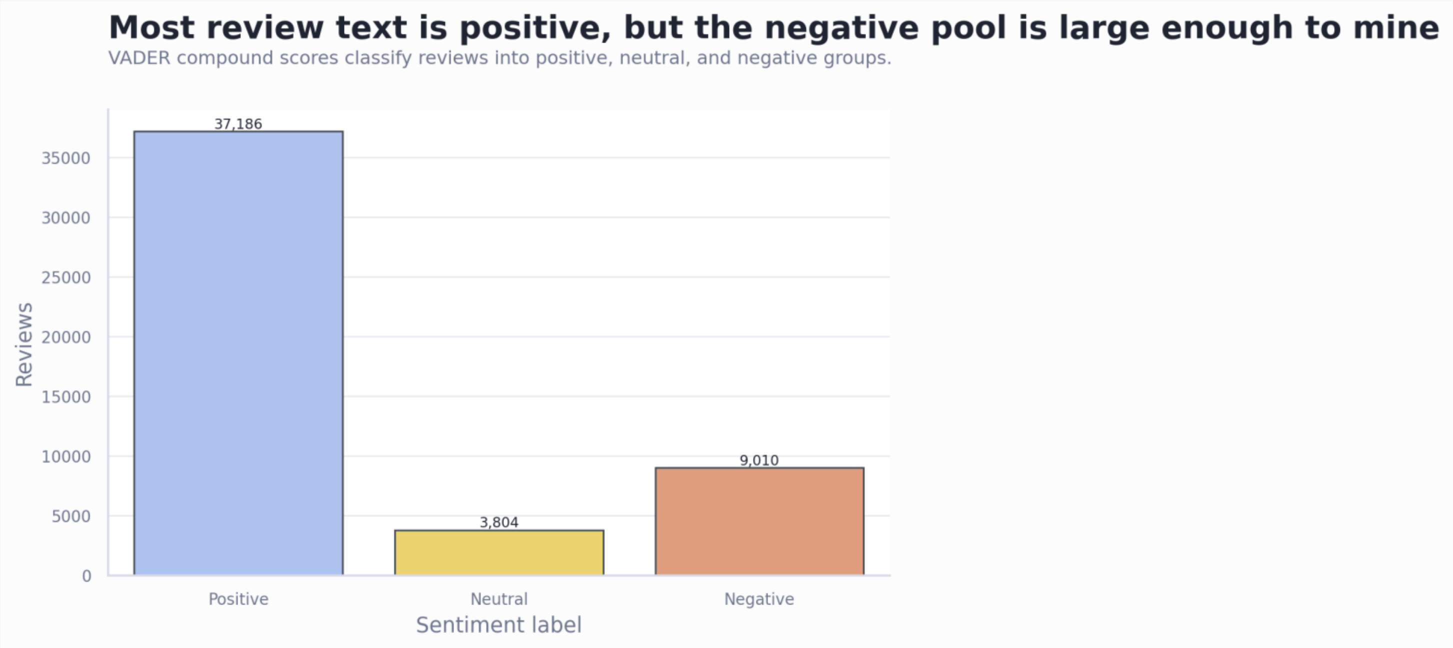
Compound score >= 0.05 is Positive, <= -0.05 is Negative, and the middle band is Neutral.

Business use

### Experience triage

Sentiment helps identify service, returns, checkout, and value-perception themes that may explain behavior.

# Sentiment: Most Reviews Are Positive, But Negative Volume Is Meaningful




**What it shows**

- Positive sentiment represents 74.4% of reviews.
- Negative sentiment is still about 18.0%, giving enough cases for issue mining.
- Repeated review templates and VADER thresholds limit precision, so human validation is recommended.

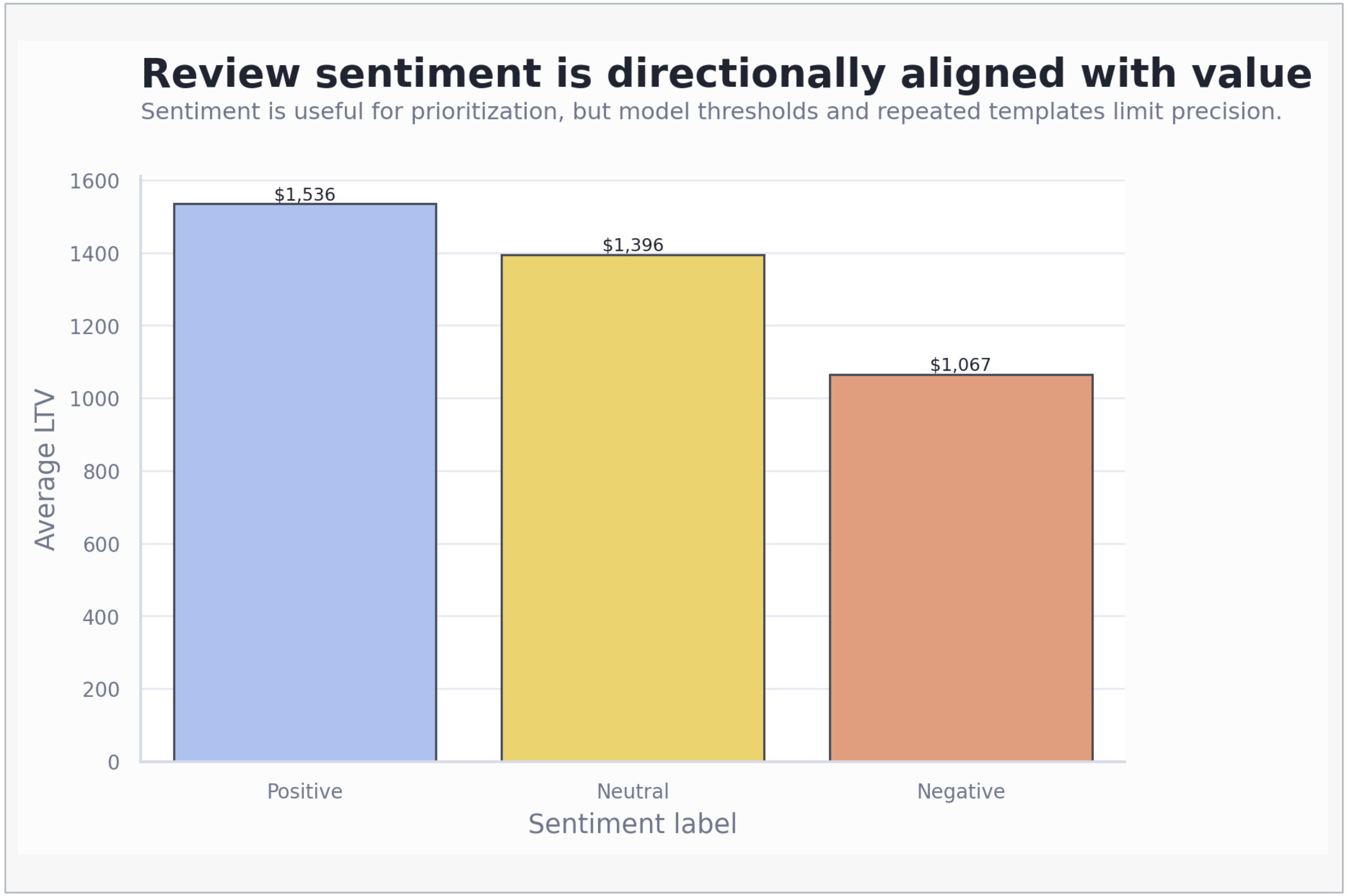
**74.4%**

Positive reviews

**18.0%**

Negative reviews

# Sentiment: Review Polarity Aligns With Customer Value




**What it shows**

- Positive sentiment has the highest average LTV; negative sentiment has the lowest.
- Sentiment should be monitored with service calls, returns, and abandonment to detect experience risk.
- This is directional evidence, not a causal estimate.

# Review Text Examples Behind Each Sentiment Class

The examples below are real review-text patterns from the dataset after VADER compound scoring. They show how the rule-based classifier maps text into Positive, Neutral, and Negative labels.

## Positive

VADER score 0.34

"I had a smooth shopping experience; the items matched the descriptions and checkout was quick. I would gladly buy from here again."

**Business reading**

Signals reliable product fit, smooth checkout, and repeat-purchase intent.

## Neutral

VADER score 0.00

"The order did not meet my expectations; service issues made the purchase feel less reliable. Overall, the experience needs work."

**Business reading**

A reminder that neutral score bands can still contain operationally negative language.

## Negative

VADER score -0.44

"I had a frustrating experience with this store; service issues made the purchase feel less reliable. Overall, the experience needs work."

**Business reading**

Points to service reliability issues that should be reviewed with support and returns data.

Use these labels as a triage layer, not a final voice-of-customer taxonomy. Topic modeling or manual coding should validate the themes before operational action.

# Insights And Findings

Insight 1

## Behavior is the strongest value signal

Total purchases, session depth, pages per session, login frequency, email open rate, and wishlist items explain more LTV variation than demographics.

Insight 2

## Cart abandonment is the main friction flag

Abandonment has a strong negative relationship with LTV, making checkout friction a priority for diagnostic work.

Insight 3

## Sentiment adds experience context

Negative reviews are large enough to mine and are directionally aligned with lower LTV, especially when paired with service and return signals.

# Recommended Actions

1

## Build behavior-led segments

Prioritize high purchase frequency, high email engagement, and high wishlist intent for personalized lifecycle campaigns.

2

## Investigate checkout friction

Analyze high-intent users with long sessions and high cart abandonment to isolate checkout, pricing, returns, or UX issues.

3

## Create a sentiment issue backlog

Cluster negative reviews by service, returns, checkout, discount dependence, and product quality themes, then link themes to LTV and churn.

# Limitations And Challenges

Data

### No customer ID or time-series history

The dataset is customer-level and does not include transaction dates, order IDs, or longitudinal events, limiting cohort and causal analysis.

Quality

### Missingness and invalid values remain risks

Median imputation and dropped fields make EDA possible, but missingness patterns and >100% rate values should be repaired upstream.

Model

### Sentiment is rule-based

VADER is transparent and fast, but it may miss domain-specific nuance, sarcasm, multilingual reviews, or template repetition.

# Next Analysis To Consider

Lifecycle

## Build customer cohorts

Compare new, growing, mature, and inactive customers by LTV, purchase frequency, email engagement, and review sentiment.

Segmentation

## Cluster behavior patterns

Create behavior-led segments using purchases, session depth, wishlist items, abandonment, service calls, and email response.

Friction

## Diagnose checkout drop-off

Profile high-intent customers who browse, save items, or open emails but still abandon carts before buying.

Measurement

## Harden data quality rules

Repair invalid rates, audit missingness drivers, add customer IDs and transaction dates, then rerun analysis at customer and order grain.

# Thank you

Questions and discussion

**E-commerce customer analysis**

Sources: analysis notebook and ecommerce customer review CSV